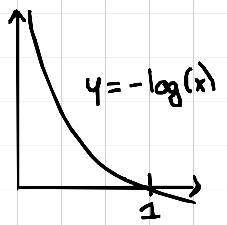
Log-Loss / Cross-Entropy Loss

For a general multi-class setting (K classes, N samples) the function is:

$$L[Y, P] = -N^{-1} \sum_{i}^{N} \sum_{i}^{K} y_{n,k} \log[p_{n,k}]$$



Where: Y matrix of one-hot encoded labels $y_{n,k}$
(1 if sample n belong to class k)
P matrix of predicted probabilities $p_{n,k}$
(where $\sum^{K} p_{n,k} = 1$)

When sample n has label k ($y_{n,k} = 1$), loss becomes $-\log(p_{n,k})$

- If model predicts $p_{n,k} = 0.99$, $-\log(0.99) \approx 0.01$ (high confidence + correct)
- If model predicts $p_{n,k} = 0.01$, $-\log(0.01) \approx 4.61$ (high confidence + wrong)

Why choose over accuracy?

- Accuracy judges predictions in binary method (correct vs. wrong), but in finance confidence of prediction can affect bet sizing.
- Using sample weighted neg. log loss accounts for more factor of prediction returns:
  - Side of bet (positive, negative)
  - Size of bet (probability of prediction)
  - Return of the bet (proportional to absolute return)